



Coleco ADAM

SMARTBASIC (APPLESOFT-COMPATIBLE) · EOS · DIGITAL DATA PACKS · Z80A · 64K

SmartBASIC is an Applesoft][work-alike — nearly every Applesoft command behaves the same. Drives: **D1/D2** data-pack tapes, **D5/D6** disks. **^X** = CTRL+key.

PROGRAM CONTROL

Applesoft-like

RUN	Start program; RUN 100
LIST	LIST 10,50 range
NEW	Erase program
CLEAR	Zero all variables
DEL 10,40	Delete line range
CONT	Resume after STOP
STOP / END	Halt program
HOME	Clear text screen

SmartBASIC is an Applesoft][work-alike, so program control matches the Apple][almost exactly.

BASIC STATEMENTS

PRINT ?	; joins, , tabs fields
INPUT GET	Read line / one key
LET	Assign (optional)
REM	Comment
GOTO GOSUB	Jump / call; RETURN
ON x GOTO	/ GOSUB a,b,c
FOR..NEXT	STEP optional
IF..THEN	IF A>5 THEN 200
DATA READ	RESTORE rewinds
DIM	DIM A(20),N\$(5)
DEF FN	DEF FN S(X)=X*X
POKE a,v	Write byte (mind EOS table!)

FUNCTIONS

ABS SGN INT	magnitude/sign/floor
SQR EXP LOG	root, e^x, ln
SIN COS TAN ATN	radians
RND(1)	random 0-1
PEEK(a)	read a byte
FRE(0)	free bytes (~26401 in 2.0)
LEN LEFT\$ RIGHT\$ MID\$	MID\$(A\$,3,2)
STR\$ VAL CHR\$ ASC	convert num/char
TAB SPC	in PRINT only

TEXT SCREEN

HOME	Clear screen, cursor home
HTAB n VTAB n	Cursor col / row
INVERSE NORMAL	Reverse / normal text
FLASH	Flashing text
SPEED=n	0 slow ... 255 fast print
TEXT	Return to text mode

LO-RES GRAPHICS

GR

GR	Enter lo-res (mixed text)
COLOR=n	Colour 0-15
PLOT x,y	Dot
HLIN a,b AT y	Horizontal line
VLIN a,b AT x	Vertical line

Palette follows Applesoft 0-15, but pixels are drawn through the ADAM's TMS9928A video chip.

HI-RES GRAPHICS

HGR

HGR HGR2	Hi-res page 1 / full screen
HCOLOR=n	Colour 0-7
HPlot x,y	Plot point
HPlot TO x,y	Line from last point
DRAW XDRAW	Shape-table draw / XOR
ROT= SCALE=	Rotation / size

ADAM DRIVES & DEVICES

the ADAM bits

D1 D2	Digital Data Pack (tape) drives
D5 D6	Disk drive 1 / 2
LOAD name,Dn	Load from a specific drive
SAVE name,Dn	Save to a drive
CATALOG,Dn	List files on drive n
FORMAT n	Format disk (5 or 6)
DELETE name	Remove a file
LOCK / UNLOCK	Protect / unprotect
PR#1	Route output to daisy-wheel printer

If no drive is given, commands use the boot device. Data packs store data in blocks, not Apple-style sectors.

SEQUENTIAL FILES

OPEN name	Open a text file
READ / WRITE	Direct I/O to open file
APPEND	Add to end of file
POSITION	Skip to a record
CLOSE	Close the file

File handling mirrors Applesoft DOS; combine with ,Dn to pick a drive.

EOS & BOOTING

Power on w/ media	Boots SmartBASIC from D1/D5
No media	Loads SmartWriter word processor
Computer Reset	Pull the reset switch to restart
OS-7 ROM	ColecoVision game compatibility
EOS	Elementary OS — device + file mgr
SmartBASIC 1.0/2.0	2.0 = newer EOS, more free RAM

Insert a cartridge and power on to use the machine as a ColecoVision.

HARDWARE & MEMORY

CPU	Zilog Z80A @ 3.58 MHz
RAM	64K (bank-switched)
Video	TMS9928A · 16K VRAM · 32×24
Sound	TI SN76489 (3 tones + noise)
PEEK/POKE	Work, but the EOS jump table moves between 1.0/2.0

KEYBOARD & DIFFERENCES

vs Apple][

Smart Keys I-VI	Context keys (mainly SmartWriter)
WP / MOVE / COPY	SmartWriter editing keys
Arrows / HOME	Cursor movement in BASIC
Video	TMS9928A ≠ Apple — hi-res geometry & colours differ
Storage	Tape/disk blocks, not Apple DOS
Printer	Bundled letter-quality daisy-wheel